

Student Guide: Using the Django Political System Prototype

1. Purpose of the prototype

This prototype is designed to help participants understand political systems as connected institutional processes rather than as lists of separate organisations. A government website usually shows visible institutions such as leaders, parliaments, ministries, laws, and public announcements. However, political outcomes are also shaped by decision-making bodies, influence networks, implementation agencies, and feedback from society.

The prototype first presents a realistic public-facing government portal. Participants then reveal the institutional architecture operating behind that interface. The main goal is to move from recognising political institutions to explaining how they interact to produce political outcomes. This matches the paper's definition of systems-level political understanding.

You do not need previous knowledge of Django or computer programming to use the prototype.

2. Learning objectives

After completing the activity, participants should be able to:

1. distinguish visible political institutions from less-visible institutional processes
2. identify decision-making institutions, influence networks, and implementation systems
3. explain how a political demand can become an authorised and implemented policy
4. trace feedback from a policy outcome back into the political system
5. compare how the same institutional layers operate differently across countries
6. explain how Django's architecture is used as an analogy for political systems

3. The four institutional layers

The prototype uses four Django terms to organise political institutions.

Django term	Political meaning	Main question
Template	Visible politics	What can citizens directly see?

View	Decision-making institutions	Who processes and authorises political decisions?
Middleware	Influence networks	Who shapes information, access, pressure, or advice?
Model	Implementation systems and institutional memory	Who converts decisions into practical outcomes?

Template Layer: The Template Layer represents the public face of politics. It includes leaders, elections, parliaments, ministries, laws, official announcements, and public services. Citizens frequently encounter this layer through government websites, news reports, speeches, elections, and public documents.

View Layer: The View Layer represents institutions that process political demands and formally authorise responses. Depending on the country, this may include legislative committees, ministries, executive offices, cabinets, or Party commissions.

Middleware Layer: The Middleware Layer represents actors that influence political decisions without necessarily possessing final legal authority. Examples include business associations, interest groups, experts, advisory councils, intelligence organisations, civil-society networks, and politically connected elites. Influence does not mean control. A group may shape how a problem is understood without being able to enact a law itself.

Model Layer: The Model Layer represents organisations responsible for implementation and institutional memory. These include government agencies, regulators, statutory boards, ministries, regional administrations, and public officials. These institutions convert authorised decisions into rules, services, funding programmes, enforcement, and administrative outcomes.

4. Activity sequence

The activity should be completed in the following order.

Step 1: Complete the pre-assessment

Answer the pre-assessment using only your existing knowledge. Do not use the software, documentation, code, or outside sources during this stage.

The pre-assessment measures your initial ability to identify and explain relationships between political institutions.

Step 2: Open the prototype

Use the link provided by the researcher. The first page allows you to select a country.

The prototype contains the following modern political cases:

- United States
- United Kingdom
- Singapore
- China
- Russia

The research paper also applies the framework to the Roman Empire, although Rome may not appear as an interactive country within the software.

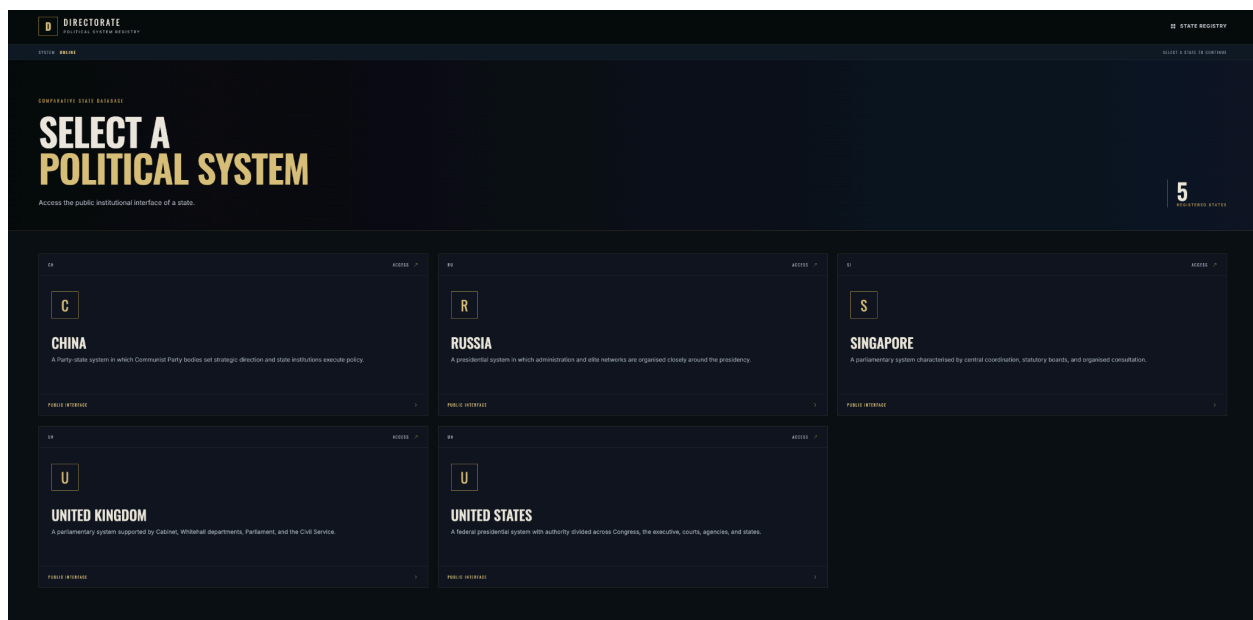


Figure 1: Country-selection interface through which participants access the comparative political cases included in the prototype.

Step 3: Select a country

Select one country to open its public-facing government portal.

Begin by examining the page as an ordinary citizen might. At this stage, focus only on what is publicly visible.

Look for:

- the country's official description;
- government institutions;
- national announcements;
- authorised or implemented policies;
- public services;
- links to government information.

5. Using the public-facing government portal

The public portal represents the **Template Layer**. It is designed to resemble an official national government website.

The portal may display the following sections:

Government information

This section introduces the country and describes its public political structure.

National announcements

This section displays policies that have reached an **authorised** or **implemented** status in the database.

A policy marked as proposed or processed will not yet appear as a public announcement.

Government institutions

This section displays formal institutions visible to citizens. These may include parliaments, ministries, executive offices, or other public organisations.

Public administration portal

The public-services panel contains links such as:

- Laws and regulations
- Government directory
- Public information
- National statistics

These links organise information already presented in the public interface.

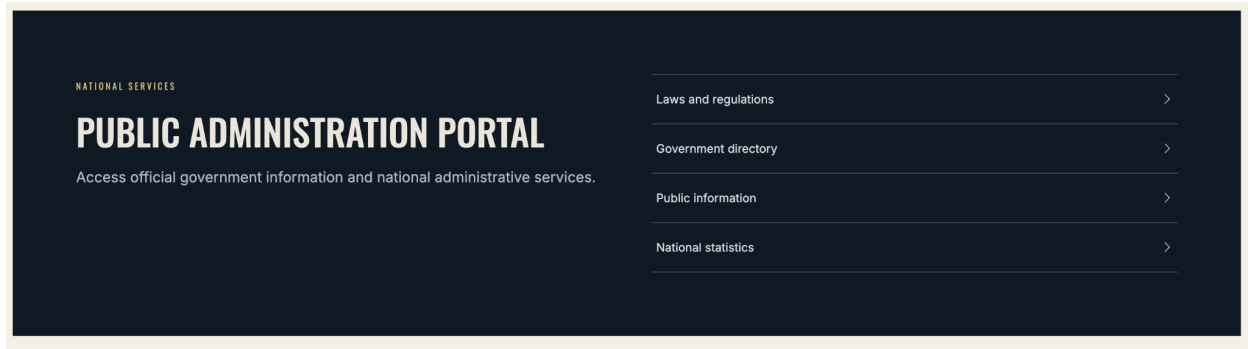


Figure 2: Public-facing country interface representing the **Template Layer**, where students initially encounter visible political institutions, official information, and published policy outputs.

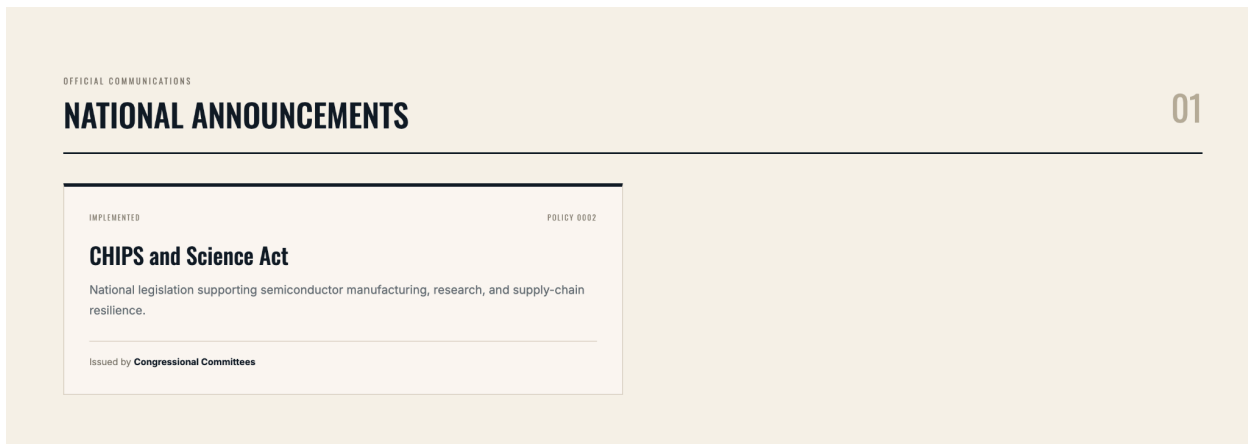
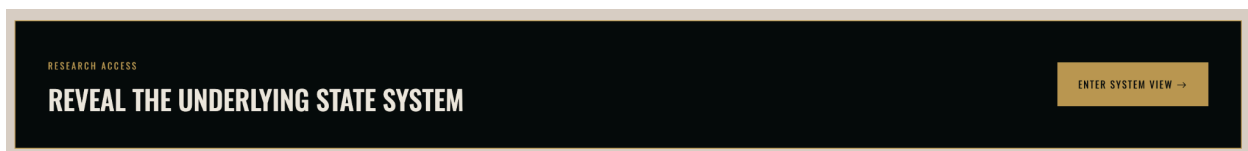


Figure 3: National announcements section displaying policies whose database status is authorised or implemented.

6. Revealing the hidden system

After exploring the public interface, select ENTER SYSTEM VIEW:



This opens the institutional architecture behind the public portal.

The system view explains how a visible policy connects with:

- a formal decision institution;
- one or more influence actors;

- an implementation institution;
- a public outcome;
- feedback.

The system view is the most important part of the activity because it shows that visible politics is produced through relationships operating behind the public interface.

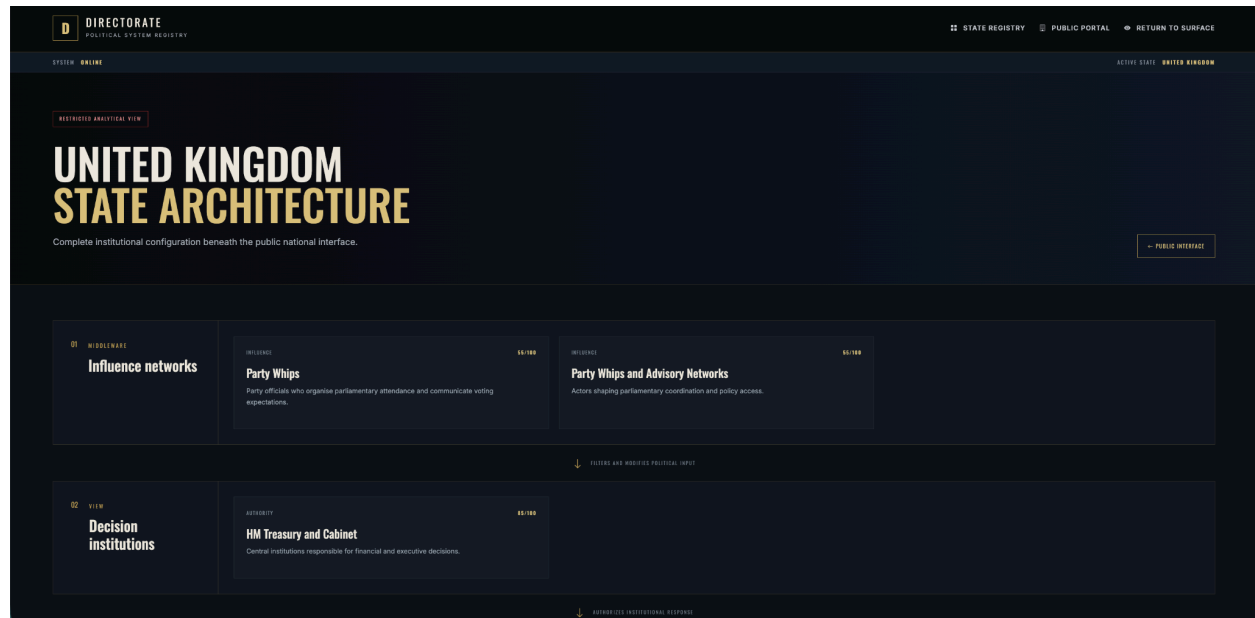


Figure 4: System view revealing the decision institutions, influence networks, implementation bodies, and policy relationships operating behind the public interface.

7. Reading the system view

Read each policy pathway in the following order.

1. Identify the public policy

Begin with the policy title and public description. This is the outcome that may appear on the public-facing portal.

2. Identify the influence actor

Ask:

- Who supplied information?
- Who provided technical expertise?
- Who applied pressure?
- Who gained access to decision-makers?
- Who shaped how the issue was presented?

This actor belongs to the **Middleware Layer**.

3. Identify the decision institution

Ask:

- Which institution processed the issue?
- Which institution had formal authority?
- Who authorised the response?

This institution belongs to the **View Layer**.

4. Identify the implementation institution

Ask:

- Which organisation converted the decision into practice?
- Who created the rules, funding process, services, or enforcement system?
- Which organisation preserved the administrative knowledge required for implementation?

This organisation belongs to the **Model Layer**.

5. Identify the public outcome

The implemented decision may appear publicly as:

- a law;
- a regulation;
- a funding programme;
- an official announcement;
- a government service;
- an administrative restriction;
- a geopolitical decision.

6. Identify feedback

Ask how the outcome returned to the political system.

Feedback may take the form of:

- public reaction;
- parliamentary scrutiny;
- judicial review;
- elections;
- media criticism;

- administrative reports;
- protests;
- expert evaluation;
- internal government reporting.

Feedback may cause the original decision to be revised.

8. Example policy pathway

A simplified policy pathway may appear as: **Influence actor** → **decision institution** → **implementation institution** → **public outcome** → **feedback**

For example, the United States semiconductor case can be understood as follows:

The Semiconductor Industry Association supplied technical knowledge and advocated for manufacturing incentives. Congressional institutions processed and authorised the CHIPS and Science Act. The CHIPS Program Office then converted the law into application requirements, funding conditions, and manufacturing awards. The public encountered the law and official funding announcements. Oversight reports later evaluated implementation and generated feedback.

The influence actor did not formally enact the law. Its role was to shape the information and pressure surrounding the decision.

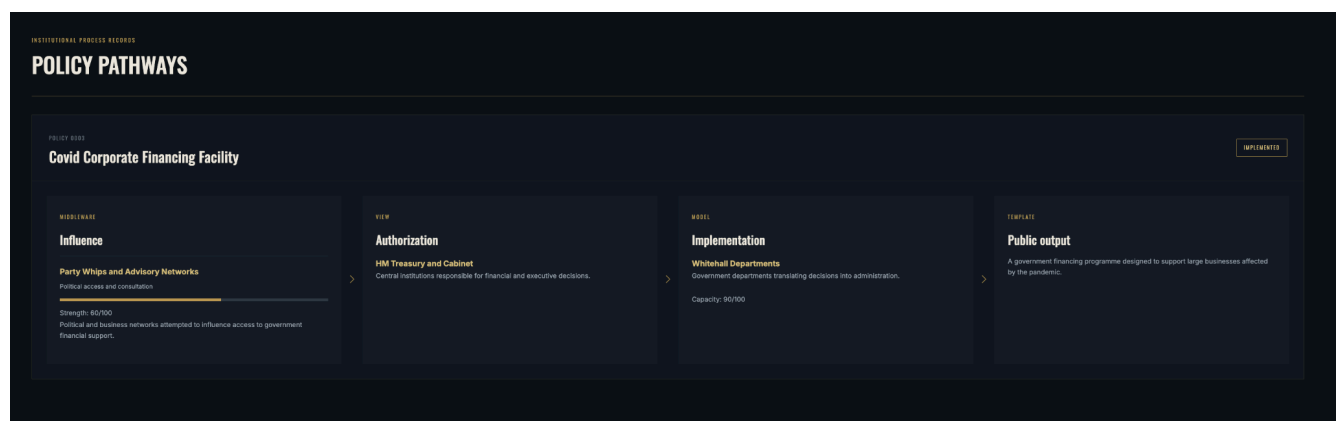


Figure 5: Example policy pathway showing how influence, formal authorisation, implementation, public output, and feedback connect across the institutional architecture.

9. Understanding the selected Django code

Participants are not expected to memorise or reproduce the code. The code is included to demonstrate how the public interface and hidden system view are produced from the same stored information.

Only short and relevant excerpts should be shown.

9.1 Template code

The HTML template controls what appears on the public page.

```
{% for policy in policies %}

    <div class="col-lg-6">

        <article class="announcement-card">

            <div class="announcement-meta">

                <span>
                    {{ policy.get_status_display|upper }}
                </span>

                <span>
                    POLICY {{ policy.id|stringformat:"04d" }}
                </span>

            </div>

            <h3>
                {{ policy.title }}
            </h3>

            <p>
                {{ policy.public_description }}
            </p>

            {% if policy.decision_institution %}
                <div class="issuing-office">
                    Issued by
                    <strong>
                        {{ policy.decision_institution.name }}
                    </strong>
                </div>
            {% endif %}

        </article>

    </div>

{% empty %}

    <div class="col-12">
        <div class="official-empty">
            No official announcements are currently published.
        </div>
    </div>

{% endfor %}
```

Code Listing 1: Django template code displaying authorised or implemented policies as public announcements.

This code means that the announcements are not permanently written into the webpage. The template displays the policy records supplied by the Django view.

9.2 View code

The Django view selects which information appears publicly and which information appears in the system view.

```
def public_state(request, country_id):
    country = get_object_or_404(Country, id=country_id)

    visible_institutions = country.institutions.filter(
        layer="decision",
    ).order_by("name")

    visible_policies = (
        country.policies
        .filter(status__in=["authorized", "implemented"])
        .select_related(
            "decision_institution",
            "implementation_institution",
        )
        .order_by("-id")
    )

    return render(
        request,
        "country/public_state.html",
        {
            "country": country,
            "institutions": visible_institutions,
            "policies": visible_policies,
        },
    )
```

```
def system_view(request, country_id):
    country = get_object_or_404(Country, id=country_id)

    decision_institutions = country.institutions.filter(
        layer="decision"
    ).order_by("name")

    influence_institutions = country.institutions.filter(
        layer="influence"
    ).order_by("name")

    implementation_institutions = country.institutions.filter(
        layer="implementation"
    ).order_by("name")

    policies = (
        country.policies
        .select_related(
            "decision_institution",
            "implementation_institution",
        )
        .prefetch_related(
            "influences__actor",
        )
        .order_by("-id")
    )

    return render(
        request,
        "country/system_view.html",
        {
            "country": country,
            "decision_institutions": decision_institutions,
            "influence_institutions": influence_institutions,
            "implementation_institutions": implementation_institutions,
            "policies": policies,
        },
    )
```

Code Listing 2 – public_state: Django view logic selecting published policies and visible institutions for the public-facing interface. A proposed or processed policy remains stored in the database but does not appear publicly until it becomes authorised or implemented.

Code Listing 3 – system_view: Django view logic separating decision, influence, and implementation institutions for the hidden system view.

The view code demonstrates the difference between the two interfaces. The public interface displays selected outputs, while the system view reveals the deeper institutional categories.

9.3 Model code

The Django models store countries, institutions, policies, and relationships.

```
class Policy(models.Model):
    STATUS_CHOICES = [
        ("proposed", "Proposed"),
        ("processed", "Processed"),
        ("authorized", "Authorized"),
        ("implemented", "Implemented"),
    ]

    country = models.ForeignKey(
        Country,
        on_delete=models.CASCADE,
        related_name="policies",
    )

    title = models.CharField(max_length=150)
    public_description = models.TextField()
    status = models.CharField(
        max_length=30,
        choices=STATUS_CHOICES,
        default="proposed",
    )

    decision_institution = models.ForeignKey(
        Institution,
        on_delete=models.SET_NULL,
        null=True,
        blank=True,
        related_name="authorized_policies",
    )

    implementation_institution = models.ForeignKey(
        Institution,
        on_delete=models.SET_NULL,
        null=True,
        blank=True,
        related_name="implemented_policies",
    )

    def __str__(self):
        return self.title
```

Code Listing 4: Django Policy model storing the country, public description, policy status, decision institution, and implementation institution.

This model stores the relationship between a policy and the institutions responsible for authorisation and implementation.

```

class InfluenceConnection(models.Model):
    actor = models.ForeignKey(
        Institution,
        on_delete=models.CASCADE,
        related_name="influence_connections",
    )

    policy = models.ForeignKey(
        Policy,
        on_delete=models.CASCADE,
        related_name="influences",
    )

    influence_type = models.CharField(max_length=100)

    influence_strength = models.PositiveIntegerField(
        default=50,
        help_text="Relative influence from 0 to 100.",
    )

    explanation = models.TextField(blank=True)

    def __str__(self):
        return f"{self.actor.name} → {self.policy.title}"

```

Code Listing 5: Django model linking an influence actor to a policy and recording the type, strength, and explanation of the relationship. This relationship allows the prototype to represent influence separately from formal legal authority.

9.4 Middleware code

The Django middleware represents the **Middleware Layer** of the political architecture. It operates between the incoming request and the final response, attaching additional political context before the page is returned to the user.

```

class PoliticalInfluenceMiddleware:
    def __init__(self, get_response):
        self.get_response = get_response

    def __call__(self, request):
        request.political_context = {
            "influence_detected": True,
            "message": (
                "This visible output has passed through "
                "an underlying influence environment."
            ),
        }

        response = self.get_response(request)
        return response

```

***Code Listing 6:** Custom Django middleware attaching political-influence context to a request before the final public response is produced.*

10. Comparing countries

The same four-layer framework is used for every country, but the institutions inside each layer differ.

United States

Decision-making is distributed across Congress, committees, the presidency, and executive agencies. Influence may enter through lobbying organisations, expert bodies, businesses, or intelligence institutions. Federal agencies implement authorised policies.

United Kingdom

Visible politics centres on Parliament, but Whitehall departments, Cabinet processes, party whips, and intelligence assessments shape how decisions reach Parliament. Civil Service organisations implement policy.

Singapore

Ministries and central institutions coordinate decision-making. Influence often enters through state-organised advisory councils, tripartite institutions, expert groups, and business consultation. Statutory boards and GovTech support implementation.

China

The public interface includes the presidency, National People's Congress, State Council, and ministries. However, major strategic direction is concentrated within Communist Party leadership bodies and central commissions. State ministries and provincial administrations implement central direction.

Russia

The presidency is highly visible, while the Presidential Executive Office processes presidential priorities. The Security Council, security-linked officials, and elite networks shape information and access. Federal ministries and regional administrations implement decisions.

The comparison demonstrates that the same visible institution can operate differently depending on its surrounding architecture.

11. Important limitations

The Django analogy is a learning model. It does not mean that politics operates exactly like computer software.

Political systems include:

- human judgement
- disagreement
- negotiation
- resistance
- informal relationships
- incomplete information
- crises
- unexpected outcomes

A political actor may ignore institutional expectations. An implementation body may reinterpret a policy. An influence network may fail to achieve its goal. Feedback may be restricted or ignored.

The framework should therefore be used as a map for examining institutional relationships, not as a deterministic explanation of political behaviour.

12. Common misunderstandings

“Middleware controls the government.”: Not necessarily. Middleware influences information, access, advice, or pressure. Formal authority remains with decision institutions.

“Once a policy is authorised, implementation is automatic.”: No. Agencies and public officials interpret rules, distribute resources, enforce decisions, and sometimes alter what citizens experience.

“Every institution fits into only one layer.”: The framework assigns institutions according to their main role within a particular pathway. In reality, an institution may perform several functions.

“The public interface is false.”: The public interface contains real political institutions and outputs. It is incomplete rather than automatically false.

“Every country has the same hidden architecture.”: No. The framework remains consistent, but the institutions and relationships differ between political systems.

13. Troubleshooting

The page does not load: Refresh the page once. If the problem continues, inform the researcher.

The announcement count displays 00: This means that the selected country has no policy currently stored with an authorised or implemented status.

The system view is empty: The country may be missing one or more institutional or policy records. Inform the researcher.

A public-service link does not work: Use the navigation menu or scroll to the relevant section, then report the broken link.

A term is unclear: Return to the four-layer table in Section 3.

14. Quick reference

Use these questions when examining any policy:

Template: What can citizens see?

View: Which institution formally processed or authorised the decision?

Middleware: Which actors shaped information, pressure, access, or advice?

Model: Which institution implemented the decision?

Outcome: What did citizens or organisations experience?

Feedback: How did the result affect later decisions?

A complete explanation may follow this sentence structure: The influence actor shaped the issue through information, access, expertise, consultation, or pressure. The decision institution

processed and authorised the response. The implementation institution converted the decision into rules, services, funding, or enforcement. The resulting outcome then produced feedback that affected later political decisions.

15. Before completing the post-assessment

Confirm that you can:

- distinguish visible politics from hidden institutional processes
- explain the difference between influence and authority
- identify decision-making and implementation institutions
- trace a policy pathway from influence to feedback
- compare how the architecture differs across countries
- explain why Django is an analogy rather than a perfect description of politics